

Student ID No.: _____

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Question #2

- A) Write the instruction needed to perform the following; where X is defined as bytes memory location ; and Y is word memory location(3 POINTS)

IF (X<5) BL=AL+X ELSE Y=BL*AL

```
CMP X,5
JNB L
MUL BL
MOV Y,AX
JMP L2
L: ADD BL,AL
ADD BL,X
L2: HLT
```

- B) Represents how **X DD 12345h,'m'** would be stored in the memory model of an Intel 8086 microprocessor system. (2 POINTS)

45H
23H
01H
00
'm' ASCII
0 (null)
0 (null)
0 (null)

- C) Write an assembler codes to test bit#3 in register BL; and perform the following operations; if bit#3 is 1 ; add 3 to Register BL otherwise clear it. (3 POINTS)

```
TEST BL,8
JZ L
ADD BL,3
JMP L2
L: MOV BL,0
HLT
```

Question #3

A) Given memory locations X and Y defined below:

```
X DB 80H,70H,50H,00H,48H,00,90H
Y DB LENGTH OF X DUP (?)
```

Write an assembler codes to perform the following (3 POINTS)

- Copy memory locations X to Y.

```
MOV CX, LENGTH OF X
MOV SI, 0
MOV DI, 0
L: MOV AL, X[SI]
MOV Y[DI], AL
INC SI
INC DI
LOOP L
```

B) Consider the following data segment memory locations, answer questions below: (2 POINTS)

0700:0102h	10	← X
0700:0103h	FF	
0700:0104h	4B	
0700:0105h	23	
0700:0106h	1F	← Y
0700:0107h	78	
0700:0108h	67	
0700:0109h	CD	

- The physical address of X = $700H \times 10H + 102H = 7102H$
- The value of DH after executing the following instruction `MOV DH, [Y-2]` is: 4BH (1 POINT)

Question #4

A) Write an assembly code to calculate the following formula and store the result at variable z then store the result in the stack. $Z = (X \times Y + 15) / 2$ (Note that X and Y byte memory locations) (3 POINTS)

```
MOV BL, 2
MOV AL, X
MUL Y
ADD AX, 15
DIV BL
PUSH AX
```

- B) Write an assembler codes to find number of locations that have values FFH ; assume you have 1 Kbyte memory locations to be tested. (3 POINTS)

```
MOV SI,0
MOV CX,1024
MOV DX,0
L2: CMP X[SI],0FFH
JNZ L
INC DX
L: INC SI
LOOP L2
```